



Outline of the lecture

The history of Informatics & HCI

Our roots and evolution

Conceptual Foundations

Information Processing | Gestalt Theory | Functional Approaches

Action Research & Praxeology

Work(place) studies & ethnography | Contextual Approaches |
Praxeology & Appropriation | Activity Theory

Structuration & Infrastructuring

A holistic look at Human-System-Interaction

HCI in Practice

Designing | Prototyping | Evaluation

The HCI Challenge

Basics

“know the user”

The HCI Challenge

Basics

Human computer interaction is a difficult endeavor with glorious rewards.

Designing interactive systems to be

effective

efficient

easy to use

enjoyable to use

*so that **people** and **society** may realize the benefits of computation-based devices*

The HCI Challenge

Basics

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Analytic
Design/Implementation

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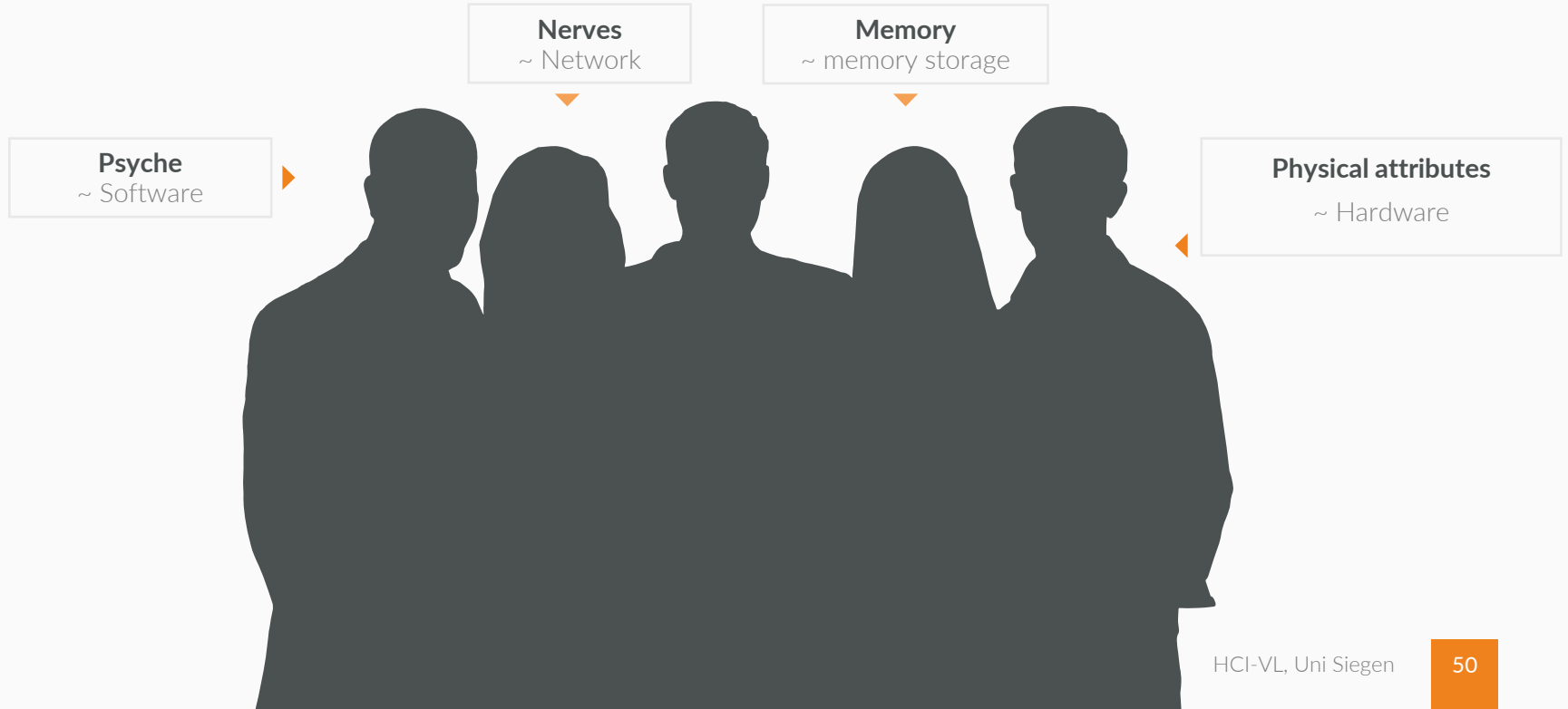
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How to go beyond platitudes like “know the user”?

Humans as information processing systems

(Questionable) Analogy: Humans as a computer-type system. Exemplary “components” are labelled:



Memory: Encoding, Storage, and Retrieval

Understanding To Design For Experiences That Last

Sensory Memory

Perception of the world/system/enviornment

Information entering through the sensory cortices of the brain and relaying through the thalamus. It lasts only milliseconds and is mostly outside conscious awareness.

Working Memory

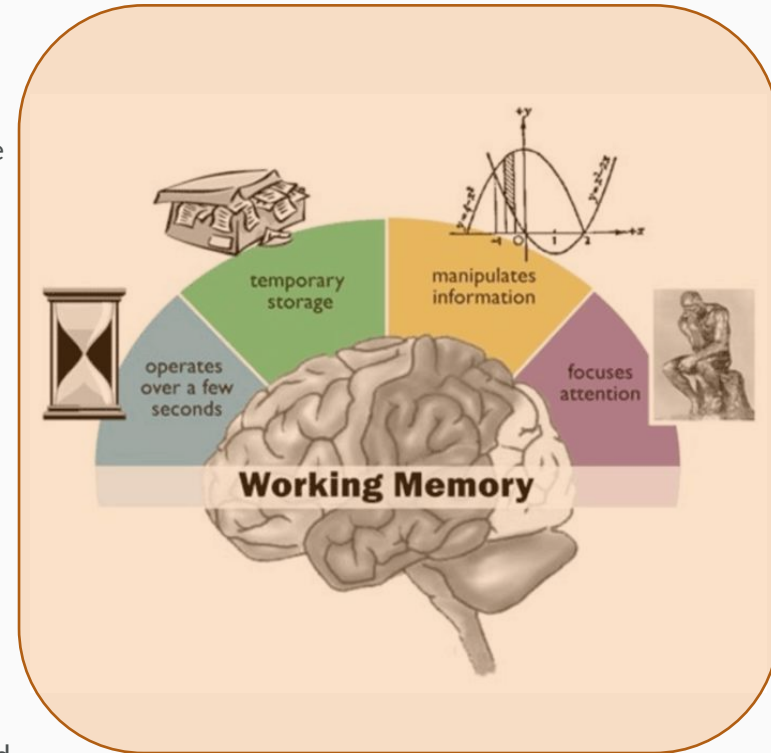
Focus of attention / Short-term memory

The structures and processes used for temporarily storing and manipulating information

Long-term Memory

Storage and Retrieval

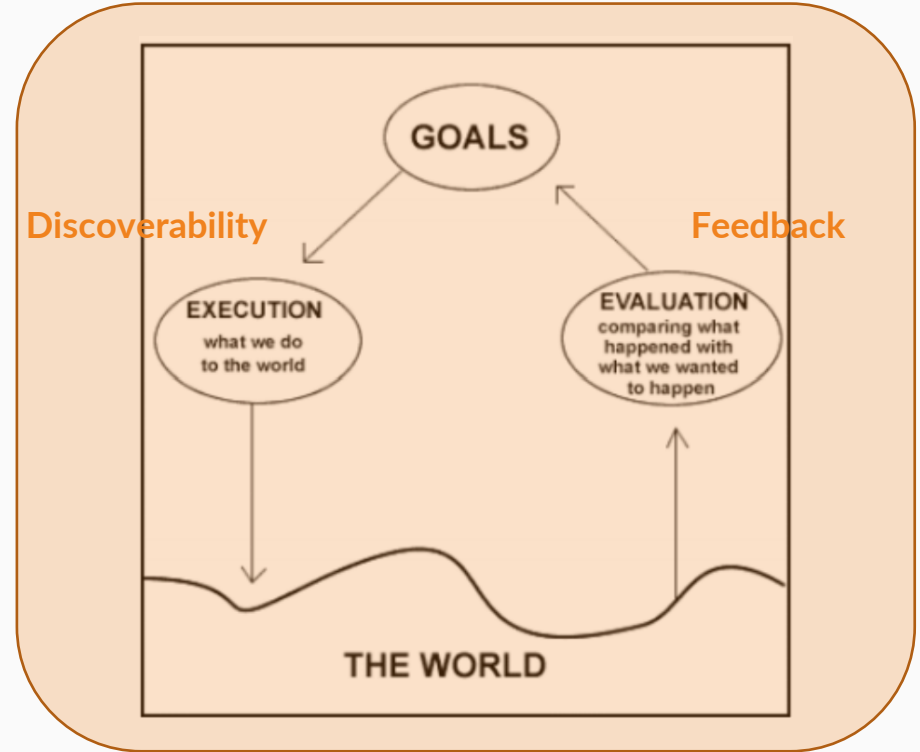
Types: Declarative, Episodic, Semantic, Implicit
Encoding: How our perceptions become memories?
Unless information is encoded, it cannot be remembered



The 7-Stage Model of Action (Norman 1988)

Predicting vs. Understanding

- Most goals not satisfied by a single action.
- Numerous sequences - discoverability
- May span seconds or minutes or hours or days.
- Continuous feedback
 - results may spawn other goals and other actions
 - goals lead to sub-goals
 - intentions lead to sub-intentions
- In a large activity, intermediate goals can be forgotten, discarded, or reformulated)

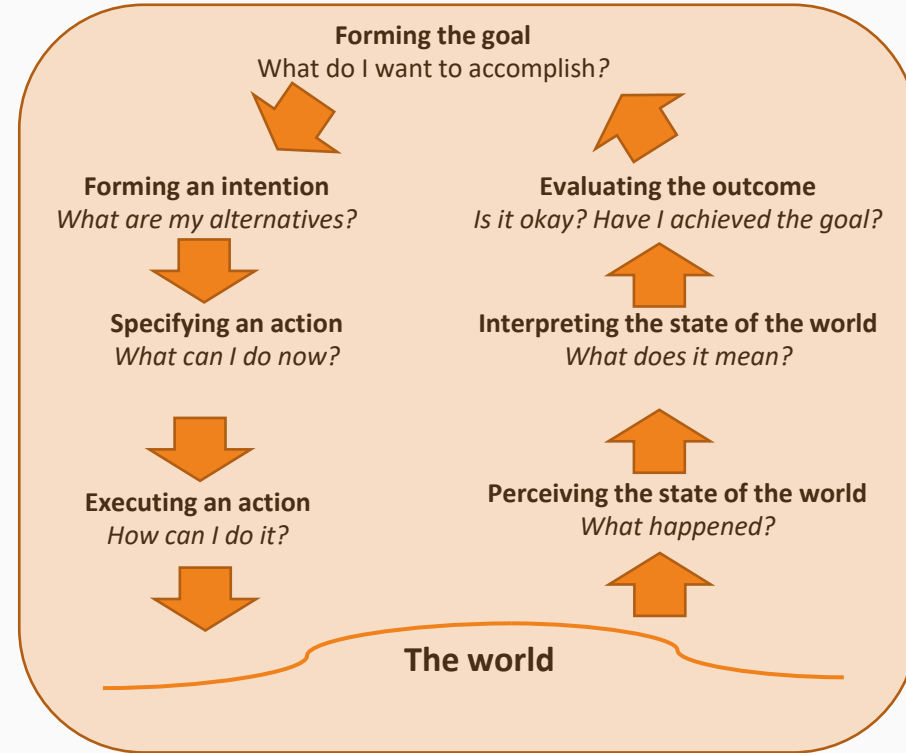


The 7-Stage Model of Action (Norman 1988)

Predicting vs. Understanding

Gulf of Execution | The gulf of execution is the difference between the intentions of the users and what the system allows them to do or how well the system supports those actions (Norman 1988)

Gulf of Evaluation | The degree to which the system/artifact provide representations that can be directly perceived and interpreted in terms of the expectations and intentions of the user (Norman 1988)

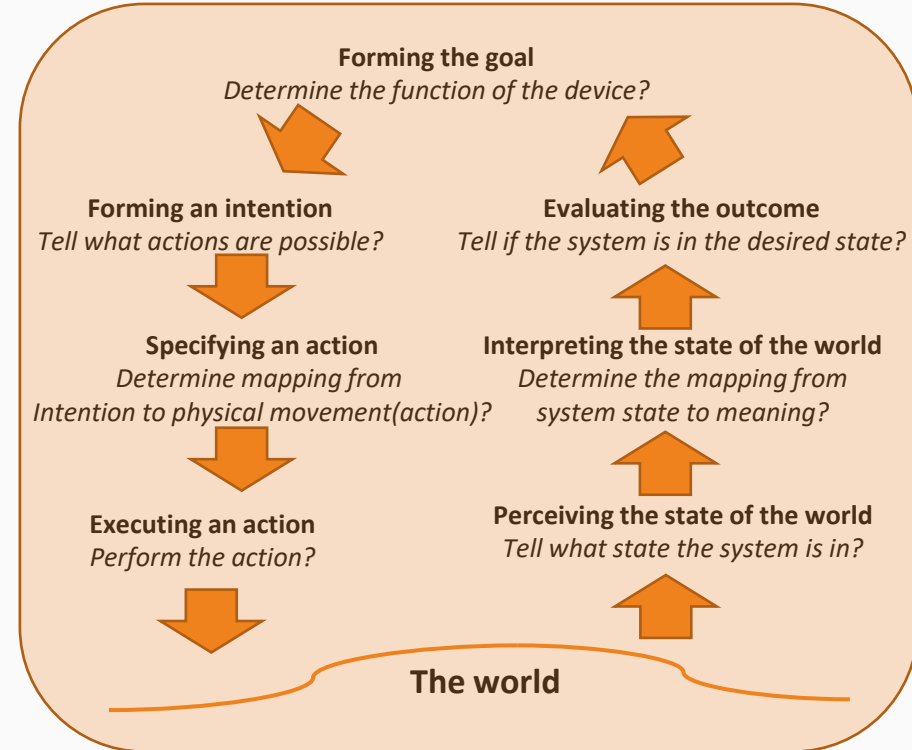


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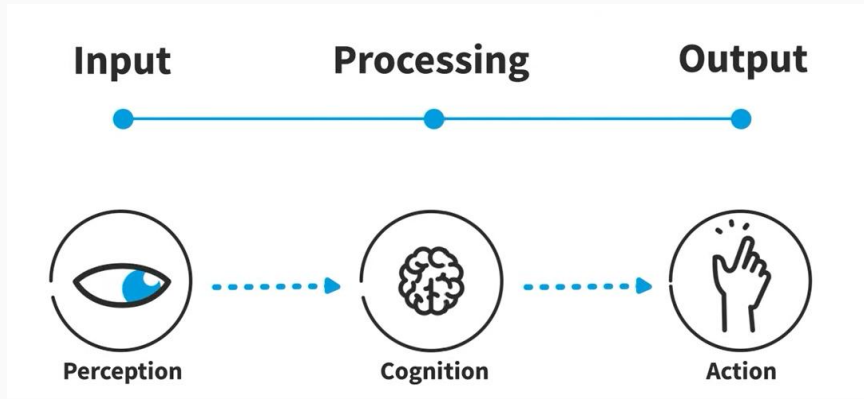
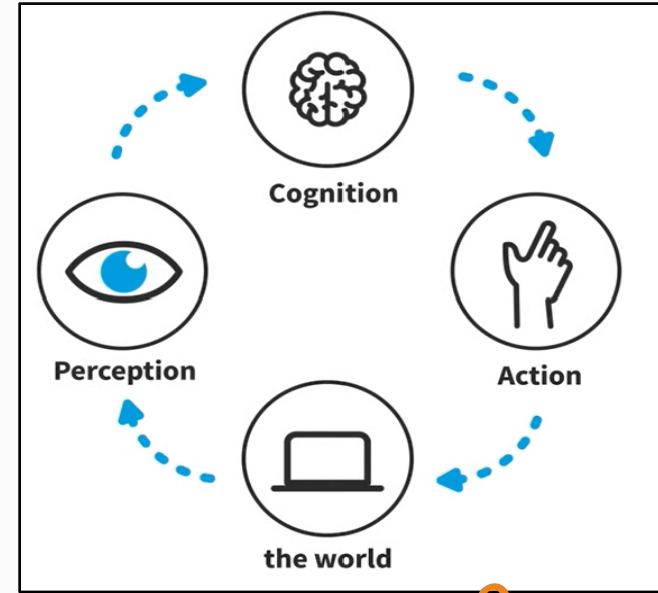


Perceptions & Illusions

Interpreting sensation

Key for constructing understanding of the world

A process through which individuals construct a mental representation of the external world, enabling them to think about it and effectively act within it. This concept extends beyond the mere reception of sensory inputs through our sense organs; it encompasses the complex interplay between cognitive processes and sensory systems to make sense of the world, especially when interacting with digital interfaces.



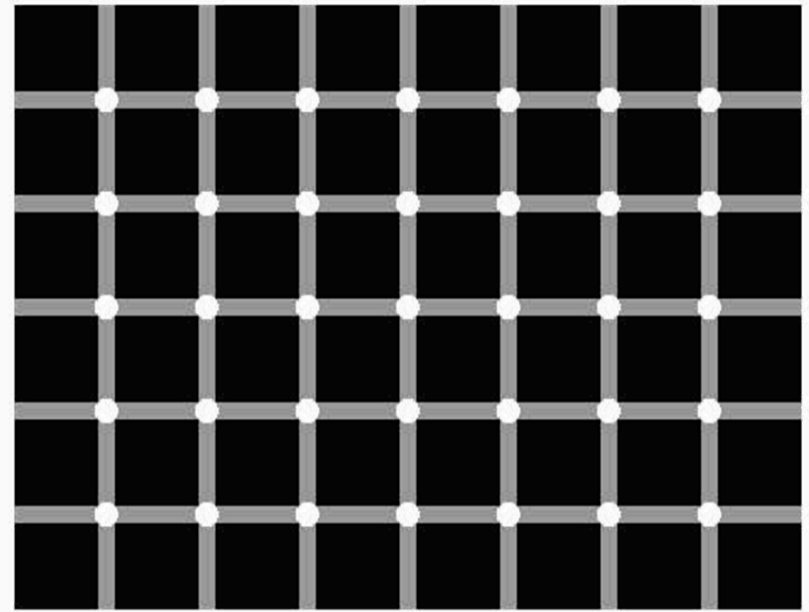
Perception vs Perspective

Perceptions & Illusions

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Illusions are special perceptual experiences in which information arising from “real” external stimuli leads to an incorrect perception, or false impression, of the object or event from which the stimulation comes



Knowledge about the physical properties of objects cannot be gained by perception, so perception is neither “veridical” nor “valid” in the strict sense of the words—the properties of the “thing in itself” remain indeterminate in any empirical sense ([Kant, 1787/1998](#))

- They tell us the limits and capacity of our perceptual apparatus
- They can specify how the constraints of perception are set



Affordances

What can a person do with the things shown here?

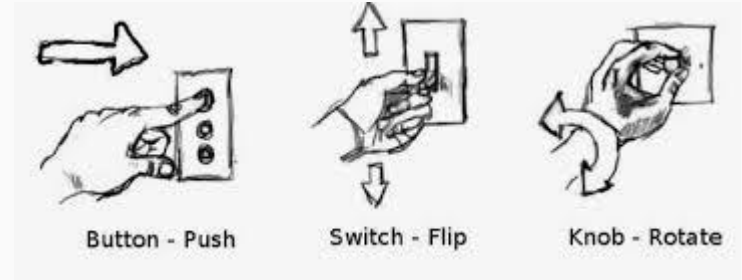


Affordance, reception & time

Why was the door designed this way?

Affordances: Expression often used unclearly

Especially in the field of usability, “affordance” is an expression often used only semi-correctly. Change in meaning over time:



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